Robust stable-roommates matchings: existence and optimization

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Abstract

For complete strict roommates preferences, a matching stable against at most d adjacent swaps in total across all lists can be found in polynomial time, or its nonexistence certified. The algorithm adds symmetry constraints to the rotation representation for robust marriage. Minimum-cost robust matching is NP-hard at every fixed swap radius. Maximizing stability probability under an explicit joint perturbation law, which need not be supported in the swap ball, is NP-hard even when that law is a point mass. We give the corresponding verification, decision and exact-search bounds.

1 Model and classification

Let I give strict complete rankings on an even set of N agents. A matching partitions the agents into pairs. An unmatched pair blocks when both agents prefer each other to their assigned partners. A matching is d-robust if it is stable after every sequence of at most d adjacent swaps across all lists, where $d \geq 0$ is encoded in binary. For the separate probability question, specify a finite joint law $\nu = \sum_{\ell=1}^{s} w_\ell \delta_{J_\ell}$ by listing complete strict profiles and nonnegative rational weights, encoded in binary, summing to one. Write

$$r_\nu(M) = \sum_{\ell=1}^{s} w_\ell \mathbf{1}\{M \text{ is stable in } J_\ell\}, \qquad M \in \mathcal{S}(I).$$

Here $\mathcal{S}(I)$ is the set of originally stable matchings. The law need not be supported in the swap ball; every listed profile is obtainable by some finite sequence of adjacent swaps. Edge costs and objective thresholds are rationals encoded in binary, and the cost of a matching is the sum of its edge costs.

Theorem 1. *For the model above, the following statements hold.*

1. *Verification, existence and construction of a d-robust matching are polynomial-time problems.*
2. *Deciding whether a d-robust matching of cost at most a given threshold exists is NP-complete for every fixed $d \geq 0$, even with zero-one edge costs. It is also NP-complete for input d.*
3. *Evaluating $r_\nu(M)$ is polynomial-time. Deciding whether some $M \in \mathcal{S}(I)$ satisfies $r_\nu(M) \geq \lambda$ is NP-complete, even for a point-mass law and $\lambda = 1$. Its maximization admits no deterministic polynomial-time multiplicative approximation with any finite factor unless $\mathsf{P} = \mathsf{NP}$.*
4. *Minimum cost subject to $M \in \mathcal{S}(I)$ and $r_\nu(M) \geq \lambda$ also has an NP-complete threshold problem. Both cost optimizations and probability maximization admit $\mathsf{FP}^{\mathsf{NP}}$ algorithms, reporting infeasibility when necessary.*

These statements concern catalogue problem C78-237 with the perturbation representation specified above. They impose no restriction on the original complete strict profile. The survey motivating the question distinguishes preference perturbations from pair-failure robustness [1, §3.3.2]; our model concerns the former.

2 Verification and a polynomial existence criterion

Number positions in agent i's original list from one, writing $r_i(j)$. For an unmatched pair put

$$b_M(i,j) = \max\{r_i(j) - r_i(M(i)), 0\} + \max\{r_j(i) - r_j(M(j)), 0\}.$$

This is exactly the minimum total number of adjacent swaps needed to make the pair block. If a name is already above the assigned partner, its contribution is zero. Otherwise reversing their order costs their position distance: they must cross each other, and each intervening name must cross one of them. Moving the candidate directly above the partner attains this bound. The two lists are distinct, so the costs add. Consequently

$$M \text{ is } d\text{-robust} \quad \Longleftrightarrow \quad b_M(i,j) > d \text{ for every unmatched pair } \{i,j\}. \tag{1}$$

This gives a quadratic verification algorithm. For $N = 0$ or 2, the unique matching is robust at every radius. For $N \geq 4$, every pair cost is at most $2(N-2)$, so $d \geq 2(N-2)$ is immediately infeasible.

We use the following established marriage representation. For a strict bipartite profile, allowing incomplete lists, in which every stable marriage is perfect, stable matchings correspond to predecessor-closed sets of rotations. With a variable x_ρ saying that rotation ρ is eliminated, predecessor closure consists of implications. Chen, Skowron and Sorge give polynomially many further implications and unit clauses whose satisfying closed sets correspond *exactly* to all d-robust stable marriages [2, Proposition 3.9, Lemmas 3.15 and 3.19, and proof of Theorem 3.24]; the form of their Lemma 3.15 used here is stated and proved below. In their notation, a relevant stable quadruple (defined below) has associated rotations π, ρ. When both exist its clause is $x_\pi \Rightarrow x_\rho$; when only π exists it is $\neg x_\pi$; when only ρ exists it is x_ρ; when neither exists the instance is infeasible. Only mutually acceptable prospective blocking pairs are considered. Thus these constraints form a polynomially constructible 2-CNF formula.

Let P_0 be a strict bipartite profile with sides U and W, possibly with incomplete lists, in which every stable marriage is perfect; a marriage pairs only agents that find each other acceptable. The swap distance $\tau(P_0, P)$ adds, over the agents, the number of pairs of names that the agent's lists in P_0 and P order differently, and it is infinite if some agent has different acceptable sets in P_0 and P. A stable marriage is d-robust if it is stable in every P with $\tau(P_0, P) \leq d$ [2, §2.1 and Definition 2.1, p. 7], and all these P have the acceptable sets of P_0. A stable quadruple $q = (u^*, w^*, u, w)$ consists of distinct $u^*, u \in U$ and $w^*, w \in W$ such that some stable marriage contains u^*w and uw^*. Its swap set $SH(P_0, q)$ consists of the adjacent swaps that move w^*, if it is below w, up to just above w in the list of u^*, and likewise u^* to just above u in the list of w^*; $P_0[SH(P_0, q)]$ is the resulting profile [2, Definition 3.10]. We call q relevant if u^* and w^* find each other acceptable and $|SH(P_0, q)| \leq d$.

A stable marriage M is then d-robust if and only if it is stable in $P_0[SH(P_0, q)]$ for every relevant q; this is the form of [2, Lemma 3.15] for relevant quadruples, and we prove it directly. Indeed, for relevant q the moves above are defined, since the list of u^* contains w^* and, as u^*w lies in a stable marriage, also w; similarly, the list of w^* contains u^* and u. Each adjacent swap reverses one pair, so $\tau(P_0, P_0[SH(P_0, q)]) \leq |SH(P_0, q)| \leq d$, and a d-robust M is stable in $P_0[SH(P_0, q)]$. Conversely, let M be stable in all these profiles, and suppose that $\{x, y\}$, with $x \in U$, blocks M in some P with $\tau(P_0, P) \leq d$. The members of a blocking pair find each other acceptable [2, §2, p. 7], in P and hence in P_0. As M is perfect, $M(x)$ and $M(y)$ exist, and

$M(x) \neq y$ because $\{x, y\}$ blocks, so $x, y, M(y), M(x)$ are distinct. Hence $q' = (x, y, M(y), M(x))$ is a stable quadruple, since M contains $xM(x)$ and $M(y)y$. With $P_1 = P_0$ and $P_2 = P$, [2, Lemma 3.14] gives $|SH(P_0, q')| \leq \tau(P_0, P) \leq d$ and shows that $\{x, y\}$ blocks M in $P_0[SH(P_0, q')]$. Thus q' is relevant and M is not stable in $P_0[SH(P_0, q')]$, a contradiction. Finally, the rotations π, ρ of a stable quadruple are unique when they exist [2, Lemma 3.18], and [2, Lemma 3.19] concerns one stable quadruple q together with $P_0[SH(P_0, q)]$. It applies to each relevant q, and its statements (iii) to (vii) show, as in the proof of [2, Theorem 3.24], that the stable marriage of a closed set is stable in $P_0[SH(P_0, q)]$ exactly when the set satisfies the clause of q given above.

Replace each original agent i by a left copy L_i and a right copy R_i. Both copies rank opposite-side names in i's original order; L_iR_i is unacceptable. Run deferred acceptance on this doubled marriage instance. If its matching is not perfect, reject: all stable marriages match the same subsets of agents, whereas a stable roommate matching would give a perfect stable marriage [2, Proposition 2.3]. Otherwise every stable marriage is perfect.

Write such a matching as a permutation p, pairing L_i with $R_{p(i)}$. The partner name of R_i is $p^{-1}(i)$. For each i and $t \in \{1, \ldots, N-1\}$ consider the two predicates

$$H^L_{i,t} = [r_i(p(i)) \leq t], \qquad H^R_{i,t} = [r_i(p^{-1}(i)) \leq t].$$

Each predicate is constant, a rotation variable, or its negation. Left partners worsen as rotations are eliminated; right partners improve. The rotations affecting an agent are ordered and traverse her stable partners without repetition. A nonconstant threshold is therefore crossed by exactly one rotation: $H^L_{i,t} = \neg x_\alpha$ or $H^R_{i,t} = x_\beta$, respectively. The crossing inequalities are $r_{\text{old}} \leq t < r_{\text{new}}$ on the left and $r_{\text{old}} > t \geq r_{\text{new}}$ on the right. The extreme stable matchings determine the constant cases, and the explicit rotations determine the crossings [2, Proposition 3.6(ii),(iv)].

Take the left copies as the side U. Every stable marriage arises from the left-optimal one by eliminating the rotations of a closed set in an order consistent with precedence, and for a closed set S this gives the stable marriage of S [2, Proposition 3.9(i) and §3.4, p. 24]. Eliminating a rotation moves each left copy in it to a later name of her list and each right copy in it to an earlier name of hers, and fixes the other agents [2, Definition 3.4]. Hence the left-optimal stable marriage gives every left copy her best and every right copy her worst stable partner; applying the same results with the sides exchanged, the right-optimal one does the opposite. So $H^L_{i,t}$ is constant exactly when it fails in the left-optimal or holds in the right-optimal stable marriage, and $H^R_{i,t}$ exactly when it holds in the left-optimal or fails in the right-optimal one. Let $H^L_{i,t}$ be nonconstant, so that it holds in the left-optimal and fails in the right-optimal stable marriage. Along an elimination order of a closed set S, the rank of the partner of L_i never decreases, and it passes from at most t to above t only when a rotation moving L_i from some R_k to some $R_{k'}$ with $r_i(k) \leq t < r_i(k')$ is eliminated. By [2, Proposition 3.6(ii)], applied to L_i and the name of rank t in her list, at most one rotation α moves L_i in this way, and α exists because the rank passes t along an elimination order that leads to the right-optimal stable marriage. If $\alpha \in S$, the rank exceeds t once α is eliminated and stays above t; if $\alpha \notin S$, it never exceeds t. Thus $H^L_{i,t} = \neg x_\alpha$ on closed sets. The ranks of right partners never increase, and the same argument with [2, Proposition 3.6(iv)] gives $H^R_{i,t} = x_\beta$.

Add $H^L_{i,t} \leftrightarrow H^R_{i,t}$ for all i, t to the marriage robustness formula. Equality of two literals, including constants, is 2-CNF. Since the copies have identical strict rankings, these equalities hold precisely when $p(i) = p^{-1}(i)$ for every i. Hence p is an involution; the forbidden diagonal excludes fixed points. Its pairs define a roommate matching.

Ordinary stability is preserved by this correspondence. Moreover, for a symmetric matching the cheapest attack by L_iR_j has cost exactly $b_M(i, j)$. Formula (1) therefore shows that robustness is preserved too: each attack concerns only the two lists of its blocking pair.

The final 2-CNF formula is satisfiable if and only if a d-robust roommate matching exists. A satisfying assignment specifies a closed rotation set; eliminate it and project its symmetric matching. Unsatisfiability certifies nonexistence. There are polynomially many rotations,

robustness clauses and threshold equalities. The preliminary radius bound handles binary d without enumerating perturbations.

3 Minimum cost is hard at every fixed radius

We first give the ordinary weighted-roommates reduction explicitly. Its four-agent gadget is the Feder construction reproduced in [3, §4.1, proof of Proposition 6]. For a Vertex Cover instance $G = (V, E)$, create p_v, a_v, b_v, c_v for each vertex v. Their preference prefixes are

$$p_v : a_v \succ \{p_u : uv \in E\} \succ c_v, \quad a_v : b_v \succ p_v,$$
$$b_v : c_v \succ a_v, \qquad\qquad c_v : p_v \succ b_v.$$

Order each neighbor block strictly and append all omitted names after the displayed prefix. Thus every list is complete.

In any stable matching, p_v must have partner a_v or c_v. Otherwise the pair $p_v a_v$ forces a_v to be paired with b_v, its only preferred alternative; then $b_v c_v$ blocks. If $p_v a_v$ is used, $b_v c_v$ must also be used, since otherwise it blocks. If $p_v c_v$ is used, $a_v b_v$ must be used for the same reason. The two possible modes are therefore

$$A_v = \{p_v a_v, b_v c_v\}, \qquad B_v = \{p_v c_v, a_v b_v\}.$$

Both are internally stable. A graph-edge pair $p_u p_v$ blocks exactly when both vertices use B. All other cross-gadget pairs are below the assigned local partners. Thus stable matchings correspond exactly to vertex covers $\{v : A_v \text{ is used}\}$. Give $p_v a_v$ cost one and every other edge cost zero. Matching cost equals cover size.

Fix any $d \geq 0$, and let this core have $n \geq 4$ agents. Choose an even integer $m > \max\{d + 1, (n-2)d\}$. Add dummies $z_0, \ldots, z_{m-1}$, with indices modulo m. Dummy z_a ranks its antipodal mate $z_{a+m/2}$ first, then the other dummies by increasing cyclic offset $1, \ldots, m-1$, omitting the mate, then all core agents. In every core list insert d distinct dummy names between each consecutive pair of core names, and append unused dummies last. This uses $(n-2)d < m$ dummies per list, with reuse across different lists.

Every stable matching pairs the mutually first-choice dummy mates. It therefore restricts to a stable core matching, and every stable core matching extends by the dummy pairs. Such an extension is d-robust. For an unmatched core pair, stability provides an endpoint that prefers its core partner; the inserted names make its attack cost at least $d + 1$. For a core–dummy pair the dummy's attack cost is at least $m - 1 > d$. For an unmatched dummy pair of offset $h \neq m/2$, the forward cost is h if $h < m/2$, and $h - 1$ otherwise. Adding the reverse cost gives $m - 1 > d$. These cases exhaust all unmatched pairs, so (1) applies. Conversely, a robust matching is stable and projects to the core.

Preserve core costs and give all dummy edges cost zero. The correspondence preserves the objective. Its size is polynomial for each fixed d, including zero. Vertex Cover hardness and the polynomial verifier prove part 2. This also proves hardness for input d; the padding itself need not be polynomial in an unbounded binary value of d.

4 Stability probability and exact search

Checking stability in all explicitly listed profiles evaluates $r_\nu(M)$ in polynomial bit time. For hardness use the NP-completeness of finding a marriage matching stable in two complete strict profiles P, Q [4, Corollary 5]. Turn both into roommates profiles by putting every opposite-side name above every same-side name. Any stable roommate matching is bipartite: a same-side pair would force a same-side pair on the other side, and one member of each pair would mutually prefer to match together. Bipartite stability is precisely marriage stability.

Take I to be the converted P, and let ν be a point mass at the converted Q. Originally stable matchings exist because complete marriage instances have stable matchings. Among them r_ν is either zero or one, and its maximum is one exactly when P, Q have a common stable matching. This proves part 3, including the approximation claim: any finite multiplicative guarantee on a positive optimum must return value one and would decide the NP-complete problem. Giving every edge cost zero also proves hardness with both a probability threshold and a cost bound. All threshold problems are in NP by the explicit verifiers.

Finally, scale rational costs, or listed probabilities, by the product of their positive denominators. This product has polynomial bit length. Each objective is then integer-valued in a range with polynomially many bits. Test feasibility first, then use binary search with NP queries to find its exact optimum. Recover an optimizer by fixing matching edges successively while retaining an optimal feasible completion; these queries remain in NP. This proves the stated $\mathsf{FP}^{\mathsf{NP}}$ upper bounds. These are polynomial-time Turing upper bounds; the decision hardness reductions above are polynomial-time many-one reductions.

Use of AI tools

The author used research AI agents that the author built and configured (GPT-6 Astra max, Claude Opus 5.5 max and GPT-6 Astra Pro) during the proof work, for proof search and construction, proof checking, translation, Python computation and similar tasks. The final manuscript was written, and the proofs were checked, by the human author. All responsibility for this work rests with the author personally.

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